

Strategic Asset Allocation

ROUND 1 · ASSET MANAGEMENT CHALLENGE 2026

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Agenda

1. Investment policy
2. Portfolio and allocation
3. Comparison with robust approaches
4. The backtest
5. Backtest protocol and robustness
6. Implementation

A long-only portfolio of six ETFs for USD 100 million: 8% target volatility, a 20% maximum drawdown tolerance, and a mix built for inflation, high rates and energy-supply risk.

Market Outlook, October 2026

- **Inflation is persistent** and interest rates are elevated
- **The Fed has tightened again**, with further policy meetings ahead
- **Long-term yields are high**, which can pressure equity valuations and bond prices
- **Energy prices are high**, adding to inflation and supply risk
- **Main risks:** persistent inflation, weaker growth and Middle East disruption
- **EEMV adds non-US equity exposure** with a minimum-volatility approach; country and currency risks remain

Sources: Federal Reserve, BLS, EIA, FRED. Detailed figures are in the report.

Objective and Risk Budget

Maximize estimated return at 8% annual volatility, subject to our diversification limits.

8.00%

VOLATILITY, BUDGET 7–9%

Last 60 monthly USD returns (Oct 2021–Sep 2026). Covariance $\times 12$; volatility $\times \sqrt{12}$.

3.80%

1-MONTH 95% VAR, USD 3.8M

$1.645\,\sigma/\sqrt{12}$ with $\sigma = 8\%$. Assumes normal returns, so real bad months can be worse: a reference, not a loss limit.

−16.19%

BACKTEST MAX DRAWDOWN

Largest fall from a peak. Our tolerance is 20%; it is not a stop-loss or a guarantee.

Self-imposed Constraints

The mandate requires long only and no leverage. We add five limits:

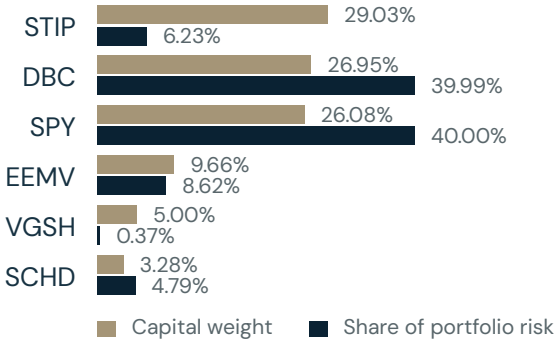
Constraint	Reason
At least **5%** in each of five blocks: US equity (SPY+SCHD), EM equity (VWO+EEMV), DBC, VGSH, STIP	Keeps exposure to every building block; not a minimum in every ETF
At least **20%** in VGSH + STIP	Retains a combined bond allocation
At most **35%** of capital in one ETF	Limits dependence on a single fund
At most **40%** of total risk from one ETF	Capital and risk weights differ, so we cap both
At most **65%** of total risk from equities	Avoids relying only on the stock market

The Portfolio: USD 100 million in six ETFs

Ticker	Fund	Weight	Risk share	Role
STIP	iShares 0–5 Year TIPS Bond	29.03%	6.23%	Inflation-linked Treasuries
DBC	Invesco DB Commodity Index	26.95%	39.99%	Broad commodity futures
SPY	SPDR S&P 500 ETF	26.08%	40.00%	US large-cap equity
EEMV	iShares MSCI EM Min Vol	9.66%	8.62%	EM equity, lower volatility
VGSH	Vanguard Short-Term Treasury	5.00%	0.37%	Short nominal Treasuries
SCHD	Schwab US Dividend Equity	3.28%	4.79%	US dividend equity
Total		**100.00%**	**100.00%**	

Constrained mean–variance optimisation, long only. VWO is eligible but receives zero weight. Risk share = $w_i(\Sigma w)_i / (w^\top \Sigma w)$.

Capital is not Risk



STIP is 29.03% of capital but only 6.23% of risk.

SPY and DBC are 53.03% of capital and 79.99% of risk.

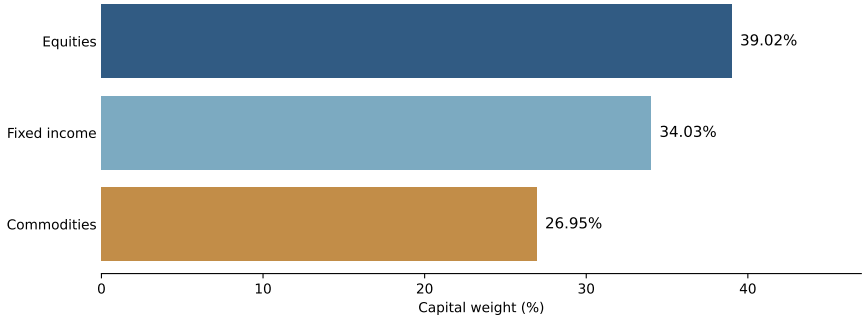
The 40% risk cap is a chosen concentration limit, not an equal-risk target.

How the Weights Were Chosen

1. **Estimate the inputs.** Use 60 monthly USD returns to estimate arithmetic mean returns, volatilities and covariances
2. **Choose all weights jointly.** Maximize estimated return at 8% annual volatility, with long-only weights summing to 100%
3. **Apply the limits.** Retain the five building blocks and bond floor; enforce the capital and risk-share caps
4. **Read the result.** SPY and DBC reach their 40% risk caps; VGSH reaches its 5% capital floor. The remaining weights balance estimated return and covariance

At the same 8% volatility and risk-free rate, maximizing estimated return also maximizes estimated Sharpe. The inputs are uncertain; this does not guarantee future performance.

Allocation by Asset Type



Capital is split across equities, commodities and short-term bonds.

Comparison with Robust Approaches

	Our MVO		Equal weight		Risk parity	
ETF	Capital	Risk	Capital	Risk	Capital	Risk
SPY	26.08%	40.00%	16.67%	27.74%	5.18%	16.67%
EEMV	9.66%	8.62%	16.67%	18.21%	7.14%	16.67%
DBC	26.95%	39.99%	16.67%	19.98%	7.85%	16.67%
VGSH	5.00%	0.37%	16.67%	1.79%	47.65%	16.67%
SCHD	3.28%	4.79%	16.67%	28.32%	5.08%	16.67%
STIP	29.03%	6.23%	16.67%	3.95%	27.09%	16.67%
Volatility	8.00%		7.80%		3.75%	

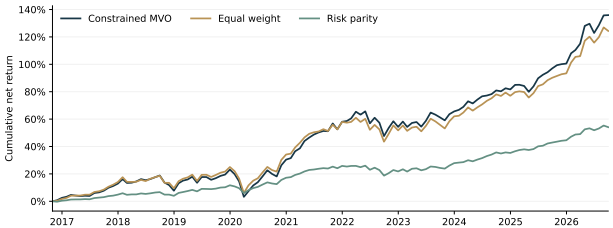
Same six ETFs, Oct 2021–Sep 2026 covariance. The comparators do not impose our MVO limits.

Equal weight: SPY and SCHD generate about 56% of the risk.

Risk parity: 47.65% in VGSH; volatility 3.75%, below the band. Scaling to 8% needs prohibited leverage.

Ours: more risk to SPY and DBC (higher means, correlation 0.119), within the 40% caps.

The Backtest: How it was run



Cumulative net returns of the fixed target allocations. Retrospective illustration.

- Nov 2016–Sep 2026, 119 monthly returns
- Fixed target weights, restored every month; no re-optimisation
- Net of 10 bp per dollar traded, including the initial investment
- Weights fitted on Oct 2021–Sep 2026, so earlier history is a retrospective illustration, not a clean test

The Backtest: Performance and Risk

9.04%	0.768	USD 235.9m
ANNUAL RETURN (MVO)	SHARPE RATIO	FINAL VALUE OF USD 100M

Fixed allocation	CAGR	Volatility	Sharpe	Max. drawdown
Constrained MVO	9.04%	8.60%	0.768	-16.19%
Equal weight	8.49%	8.30%	0.731	-15.17%
Risk parity	4.46%	3.53%	0.561	-5.81%

The MVO Sharpe is only modestly above equal weight. Risk parity lost less but sits below the required volatility band.

Backtest Protocol

- **Design:** fixed target weights restored monthly, Nov 2016–Sep 2026.
- **Data used to choose:** Oct 2021–Sep 2026 returns (60 months).
- **Data held out:** none. History was also reviewed when choosing ETFs, limits and methods; fits before Nov 2016 or Nov 2021 are conditional checks only.
- **Variations tried:** 17 method labels, 18 diversification subsets, 255-subset screens; then 108 constraint settings, 32 return-input scenarios, 33 volatility ceilings. Counts overlap. Earlier trials informed selection; later robustness checks did not change the final allocation.

Realised volatility	8.60%
Max drawdown	-16.19%
Historical VaR (5%)	3.68%
Backtested Sharpe	0.768
Discounted Sharpe	**0.384**

50% judgmental haircut: many alternatives were inspected, the fitting window overlaps the evaluation period, and expected returns are uncertain.

Robustness and Stress Tests

Risk caps of 37.5% / 42.5% reallocate 3.64% / 3.83% of capital; 35% / 45% reallocate 7.10% / 11.76%.

VGSH annual mean +1 pp: 28.84% of capital reallocated. Under the original inputs, annual expected return is 6.38 bp lower. This shows sensitivity to return estimates.

Scenario	Return
2018 Q4	-9.35%
Feb–Mar 2020	-13.82%
2022	-2.35%
Growth shock	-11.89%
Commodity reversal	-12.05%
Inflation / rates shock	-2.49%

Gross, selected weights. Periods differ in length; shocks have no assigned probability.

Expected shortfall (5%): 5.67%, USD 5.67m, against 3.68% VaR.

Rolling 12-month volatility ranged from 3.04% to 16.51%.

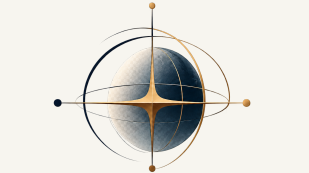
SPY and DBC produced USD 57.61m and 49.13m of the USD 135.90m net profit.

Implementation

Bloomberg entry: 16 Oct 2026; prices: 16 Oct close (USD). Capital: USD 100m.

Ticker	ETF	# shares	Price	Weight
SPY	State Street SPDR S&P 500 ETF Trust	34,169	763.25	26.08%
EEMV	iShares MSCI Emerging Markets Min Vol Factor ETF	133,610	72.30	9.66%
DBC	Invesco DB Commodity Index Tracking Fund	798,756	33.74	26.95%
VGSH	Vanguard Short-Term Treasury Index Fund ETF	86,671	57.69	5.00%
SCHD	Schwab U.S. Dividend Equity ETF	95,488	34.35	3.28%
STIP	iShares 0–5 Year TIPS Bond ETF	288,630	100.58	29.03%
Total				**100.00%**

Opening market value USD 99,999,987.88; remaining cash USD 12.12. Shares are whole numbers, so actual weights differ slightly from the targets.



Thank you